

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Electronics and Communication Engineering
End Semester Examination, Jan-June 2023

B.Tech(ECE): Semester-II

Course Name: Computer Hardware and Networking

Maximum Time: 3 hours

Course Code: CS25102

Max. Marks: 60

Instruction:

1. Attempt any six questions.
2. Assume any suitable data, if necessary.
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

		Marks	CO	BL
1	What are different storage devices in Computer system? Explain them and where and when they are used?	10	CO1	Understand and Apply
2	Explain operating system and functions of operating system. What is cpu scheduling and why it is needed. Five processes are there: p1, p2, p3, p4, p5 arrived at time 1, 2, 3, 4, 5 and burst time required are 7, 5, 1, 2, 8 respectively. What is the average turnaround time and average waiting time if the shortest job first with non preemptive is used.	10	CO2	Understand and Apply
3	Explain OSI model and its layers in computer networks. What is Hamming code error detection and correction? Suppose you have a message 11011, find parity bits. How error is determined.	10	CO3	Understand and Apply
4	What is the name of a data unit at the physical, data link, network layer, transport layer? What are the various functions of the network layer? How many spanning trees are possible for a network of 100 computers where each computer is connected with all other computers in the network. How many spanning trees are possible for the given network.	10	CO3	Understand and Apply
5	Explain IPv4 addressing. What are various classes and their range? Given Ip address is 194.45.34.28. Find its network id and broadcast address.	10	CO3	Understand and Apply
6	Explain network topologies. What is mesh topology? How many wires are required if you have 100 computers in a mesh topology. Explain transmission modes with examples.	10	CO3	Understand and evaluate
7	Explain wired and wireless communication media. Explain networking devices and where they are used with respect to OSI layers.	10	CO4	Understand and analyze

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Computer Science and Engineering
MID SEMESTER EXAMINATION, Jan-June 2023

B.Tech(ECE): Semester-II

Course Name: Computer Hardware and Networking

Maximum Time: 2 hours

Course Code: CS25102

Max. Marks: 30

Instruction:

1. Attempt all questions.
2. Assume any suitable data, if necessary.
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

		Marks	CO	BL
1	Differentiate between RAM and ROM.	6	CO4	Understand and apply
2	Define cache memory. Explain the hard disk. What are hard disk <u>sectors</u> , <u>clusters</u> , track and cylinder.	9	CO4	Understand and Apply
3	What is a process? Explain the states of a process with a state transition diagram.	5	CO2	Understand and Apply
4	What is context of a process? What is context switch. Explain process control block?	4	CO5	Analyze and Evaluate
5	Consider <u>three processes</u> whose <u>burst time</u> is 10, 20, 30 time units and arrive at time 0, 2 and 6 respectively. Assume preemptive <u>shortest job first</u> is followed, find the <u>average turnaround time</u> and <u>average waiting time</u> .	6	CO2	Understand and Apply

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